

STA 314: Statistical Methods for Machine Learning I

General Information

Instructor: Tatiana Krikella

Email: sta314@course.utoronto.ca

- Use the course email to discuss personal issues where Piazza (see Course Materials and Important Links section) is not an appropriate way to discuss. **Instructor and TAs will not respond to emails sent to individual accounts.**

Response: It will take 24 – 48 hours to respond to emails. I will only respond during working hours (Monday – Friday, 9am – 5pm).

Course Description

Machine learning (ML) is a set of techniques that allow computers to learn from data and past experience, rather than requiring humans to specify the desired behaviour by hand. ML has become increasingly central both in statistics as an academic discipline, and in the data science industry. This course provides a broad introduction to commonly used ML methods, as well as the key statistical concepts underlying ML. It serves as a foundation for more advanced courses, such as STA414 (Statistical Methods for Machine Learning II).

We will cover popular statistical methods for supervised and unsupervised learning from data as well as important concepts used in the field, including: training error, test error and cross-validation; classification, regression, and logistic regression; variable selection; penalized regression; principal components analysis; stochastic gradient descent; decision trees and random forests; neural nets; k-means clustering and nearest neighbour methods. Computational tutorials will support effective application of these methods.

Prerequisites

- **Programming basics:** CSC108H1 / CSC110Y1 / CSC120H1 / CSC148H1 / CSCA08H3 / CSCA48H3 / CSCA20H3 / CSC108H5 / CSC148H5
- **Multivariate calculus:** MAT235Y1 / MAT237Y1 / MAT257Y1 / (MATB41H3, MATB42H3) / (MAT232H5, MAT236H5) / (MAT233H5, MAT236H5)
- **Linear algebra:** MAT223H1 / MAT224H1 / MAT240H1 / MATA22H3 / MATA23H3 / MAT223H5 / MAT240H5 / MATB24H3 / MAT224H5
- **Statistics:** STA302H1 / STA302H5 / STAC67H3

Exclusions

- CSC311H1, CSC311H5, STA314H5, STA315H5, CSCC11H3

Course Materials and Important Links

Course Website

All date information, location information, and most of the course materials (schedule, lecture and tutorial slides, readings, assignments) can be found on the [course website](#).

Quercus

We will make announcements and post non-public materials, e.g. assignment and test solutions, on [Quercus](#).

Piazza

We will use Piazza for the course forum. If your question is about the course material and doesn't give away any hints for the assignment, please post to Piazza so that the entire class can benefit from the answer. You can access Piazza through the Quercus website.

Crowdmark

We will use Crowdmark for assignment submission on Quercus.

Statistical Software

Statistical programming is an essential part of this course. We will use R throughout this course, which is a free, open-source statistical programming language widely used for data analysis and reproducible research.

The software is publicly available for free at <https://mirror.csclub.uwaterloo.ca/CRAN/>, and it is also highly recommended to install RStudio (<https://posit.co/downloads/>) for an interactive programming environment. Note that the University of Toronto offers an open-source web application to use R and RStudio: <https://r.datatools.utoronto.ca/>.

Textbooks

We will mainly use the following textbook for the course

- Gareth James, Daniela Witten, Trevor Hastie, and Robert Tibshirani. An Introduction to Statistical Learning. <https://www.statlearning.com>

Students are only responsible for the material covered in lectures, tutorials, and assignments. There are many other publicly available references that you may find useful, such as

- Trevor Hastie, Robert Tibshirani, and Jerome Friedman. The Elements of Statistical Learning.
- Chris Bishop. Pattern Recognition and Machine Learning.
- Kevin Murphy. Machine Learning: a Probabilistic Perspective

Delivery Details

Unless otherwise specified, lectures and tutorials will be held in person. There will be no synchronous online video stream or recordings of the lectures. Students should be enrolled in a lecture section and a tutorial section. Tutorial sessions complement the lectures by reviewing and extending important concepts and methods, as well as providing demonstrations of computational techniques used for statistical analysis. Students are strongly encouraged to attend all lectures and tutorials. The first lecture will be on Wednesday, September 9th. There will be no tutorials in the first week of the class.

Section	Category	Time	Location
LEC0101	Lectures	Mon (10–11am)	PB B250
		Wed (11am–1pm)	SF 1105
	Instructor's Office Hours Tutorials (101–104)	Mon (1pm–2pm)	Online, see Quercus for link
		Mon (11am–12pm)	See ACORN

Course Grading Scheme

Your final course grade will be calculated using the grading scheme that results in the higher overall mark. The standard grading scheme and the flexible grading scheme are shown below.

Assessment	Standard Scheme	Flexible Scheme
Assignment 1	10%	10%
Assignment 2	10%	10%
Midterm Test	30%	15%
Final Exam	50%	65%

The flexible grading scheme is intended to recognize improvement over the course of the semester by shifting some of the weight from the midterm test to the final exam when a student demonstrates stronger performance later in the course. The grading scheme that results in the higher overall course grade will be applied automatically; students do not need to request this adjustment.

Assignments

Assessment	Date Released	Due Date
Assignment 1	Sept. 23	Oct. 7
Assignment 2	Nov. 18	Dec. 2

Format

All assignments must be completed by hand (digital handwriting using a tablet, such as an iPad, is also acceptable), unless otherwise specified in the assignment instructions. Please ensure that all submitted work is clear and legible. Assignments must be submitted as a single PDF file through Crowdmark.

Lateness

Assignments may be submitted up to 2 days late without penalty. After this 2-day grace period, a penalty of 10% of the total assignment grade per day will be applied, with partial days rounded up to the nearest full day. Assignments will be accepted for a maximum of 5 days after the original deadline. Assignments submitted more than 5 days late will receive no credit.

Requests for extensions beyond the two-day grace period due to illness, injury, or other legitimate circumstances should be submitted using the Academic Consideration Form (see the Absence Declaration section). If you expect to miss more than one week of class due to illness or an emergency, please contact your college registrar's office as soon as possible to notify them of your circumstances.

Collaboration Policy

You are welcome to work with other students and use outside resources (e.g., textbooks, papers, and online tutorials) to help you understand the assignment material. However, your submitted solutions must be written independently. You must cite any collaborators and outside resources used. See the Policy on Generative AI section for permitted uses of GenAI tools.

Tests

The course includes one midterm test and one final examination, each with a duration of 2 hours. The midterm test is held during the normal class time while the final exam is held in the final assessment period (see the dates and locations below). All tests will be closed-book. Students are responsible for the material covered in lectures, tutorials, and assignments. More details on the tests will be provided during the term.

Assessment	Date and Time	Location
Midterm Test	November 4, 11am – 1pm	EX 200
Final Exam	During final exam period, date to be announced	TBD

Missed Tests

If you miss the midterm for a legitimate reason, you must submit a request for academic consideration using the Academic Consideration Form (see the Absence Declaration section). If approved, the 30% weight of the midterm will be transferred to the final exam. In this case, the final exam will be worth 80% of your final course grade.

Collaboration Policy

Collaboration on the tests is strictly prohibited, and you may not discuss the test with anyone other than the instructor or TAs. Each student is responsible for their own work. Violation of this policy is an academic offence and will be investigated and reported as such.

Remark Requests

To request a remark, fill out this [Microsoft Form](#), where you will indicate the assessment, the question you would like regraded, and provide a justification that refers specifically to your answer and the relevant course material. Requests without this justification will not be considered. Requests will be considered by the same TA who marked the assessment. The deadline for requesting a remark is one week after the marked assessment is returned. A regrade may result in the mark increasing, decreasing, or remaining unchanged.

Note that all final exam requests go through FAS. For more information, please visit [Exam Reread & Course Mark Recheck](#)

Absence Declaration

If illness, injury, an emergency, or another legitimate circumstance affects your ability to complete academic work, please request academic consideration as soon as possible using the procedure below.

If documentation is required to support a request for academic consideration, accepted forms include an Absence Declaration through ACORN, a Verification of Illness or Injury (VOI) form, a College Registrar's letter, or a Letter of Academic Accommodation from Accessibility Services. One form of recognized documentation is sufficient.

To request academic consideration, please complete the course [Academic Consideration Form](#) and upload the appropriate documentation or proof of your ACORN Absence Declaration. Requests should not be submitted by email.

Once the form has been submitted, the instructor will review the request and determine the appropriate academic consideration (e.g., an extension or reweighting of a missed assessment) in accordance with the course policies.

For extended absences or other circumstances affecting multiple courses, students are encouraged to contact their College Registrar's Office.

Academic Integrity

The University supports acting in honesty, trust, fairness, respect, responsibility, and courage in all academic matters. Students are responsible for knowing the content of the University's Code of Behaviour on Academic Matters. All suspected cases of academic dishonesty will be investigated following procedures outlined in the Code of Behaviour above. If you have questions or concerns about what constitutes appropriate academic behaviour or appropriate research and citation methods, you are expected to seek out additional information on academic integrity from your instructor or from other institutional resources (<http://academicintegrity.utoronto.ca/>).

Policies on Generative AI

Generative AI (GenAI) tools may be used as **learning aids** in this course. For example, you may use GenAI to ask general questions about course concepts, request additional explanations, or

help you study. Keep in mind that GenAI tools can produce incorrect or misleading information, and you are responsible for verifying any information you use.

For assignments, you may use GenAI to ask general conceptual questions, but you may not provide assignment questions (in whole or in part) to a GenAI tool or use GenAI to generate, solve, provide hints for, or write any part of your assignment solution. Unless otherwise stated, all submitted work must be your own.

The use of GenAI is not permitted during tests or examinations.

Campus Resources

Accommodations

If you have a consideration that may require accommodations, please contact Accessibility Services at (<https://studentlife.utoronto.ca/as>) or (416) 978-8060. For students being supported by Accessibility Services, it is recommended (though not required) to keep the instructor updated so that individualized assistance or accommodations (beyond the ones offered by AS) can be applied if needed. Students are not required to disclose private medical information or details of a disability to the instructor.

Learning Skills

The Centre for Learning Strategy Support offers workshops, one-on-one consultations, peer mentoring, and other resources to help students develop effective study strategies and academic skills. Writing Centres are also available to support academic writing and communication. More information can be found on their [website](#).

Health Resources

Health & Wellness provides physical and mental health services, counselling, and wellness programs. Students are encouraged to seek support early if they are experiencing academic, personal, or mental health challenges. More information can be found on their [website](#).